DSA210 Final Report

Project title:

The Relationship Between Menstrual Cycle and Study Focus of Female Students

Name, Surname & ID:

Beril Nur Serbest – 22630

Supervised by:

Lecturer Onur Varol and Teaching Assistant Özgün Yargı

Date:

9 January 2026

Introduction

Understanding factors that influence academic productivity is an important topic in data science, particularly when behavioral and physiological variables interact. Among these factors, the menstrual cycle is known to affect physical condition, mood, and cognitive performance due to hormonal fluctuations and menstrual-related symptoms. However, its relationship with daily study behavior has not been extensively explored using data-driven approaches.

This project investigates the relationship between menstrual cycle phases and study focus of female students. By analyzing self-tracked daily study logs alongside menstrual cycle data, the study aims to examine whether menstrual period days are associated with changes in study duration. In addition to menstrual cycle status, external behavioral factors such as exam periods, coffee consumption, and painkiller usage are also considered to provide a more comprehensive analysis.

Motivation

The motivation behind this project is twofold. First, as a female university student, managing an effective and sustainable study routine is particularly important, as academic work constitutes a significant part of daily life. Menstruation is an unavoidable biological reality for women, and its potential effects on energy levels, focus, and well-being cannot be separated from everyday academic responsibilities. Consequently, study habits and academic planning cannot be considered entirely independent of menstrual cycle phases. This personal need to understand and balance physiological factors with academic demands served as a primary motivation for selecting this research topic.

Second, given the large number of female students in higher education, this study aims to contribute to broader awareness by providing data-driven insights that may be relevant to many individuals beyond the scope of this project. By approaching menstrual cycle effects through quantitative analysis rather than subjective assumptions, the project seeks to promote a more informed and realistic understanding of women's study behavior. In this sense, the study not only addresses a personal academic concern but also aspires to offer findings that may be beneficial for a wider female student population.

Data Source and Data Collection

This study is based on self-collected and self-tracked data, where the author of this study served as the sole participant. The research was designed as a single-subject longitudinal analysis, aiming to examine daily study behavior alongside menstrual cycle information over time.

Daily behavioral data were recorded by the author through Google Forms. On a daily basis, the author logged her study duration (in hours) and reported whether the day coincided with an exam period, whether she consumed coffee, and whether she used painkillers. These variables were included to account for external and behavioral factors that could influence study focus independently of menstrual cycle status.

Menstrual cycle data were collected using the Beije menstrual cycle tracking application, which was used to record the start and end dates of menstrual cycles. Based on these records, menstrual period days were identified, and the length of the menstrual period was calculated accordingly. This information enabled the classification of each day as a menstrual or non-menstrual day in the dataset.

Data were collected for at least one complete menstrual cycle, ideally spanning four to six weeks, with each observation representing one day. This resulted in a time-series dataset reflecting the author's real-world study behavior over consecutive days.

All data were collected voluntarily and used solely for academic purposes. The dataset was anonymized prior to analysis, and no personally identifiable information was included.

Data Preprocessing

Data preprocessing was conducted to ensure data quality, consistency, and suitability for statistical analysis and machine learning methods. Since the dataset was self-tracked over time, particular attention was given to handling potential inconsistencies and preparing variables in a format appropriate for quantitative analysis.

Data Cleaning

The raw data were examined for missing, inconsistent, or extreme values. Days with incomplete records were excluded, and all date formats were standardized to ensure chronological consistency. Variable names were checked for consistency, and minor input inconsistencies were corrected to maintain uniformity across the dataset.

Feature Engineering and Transformations

Several transformations were applied to prepare the dataset for analysis. Boolean variables, including painkiller usage, exam period, and coffee consumption, were converted into numerical format (0 = No, 1 = Yes). Based on menstrual cycle tracking data, each day was labeled as either a menstrual period day or a non-menstrual day, resulting in a binary variable indicating menstrual status.

Additionally, the dataset was structured so that each row represented a single day, preserving the temporal nature of the observations. This structure enabled both time-series analysis and time-aware machine learning evaluation.

Final Dataset Preparation

After preprocessing, all cleaned and transformed variables were combined into a single finalized dataset. The resulting dataset was free of missing values and fully prepared for exploratory data analysis, hypothesis testing, and supervised machine learning modeling.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to gain an initial understanding of the dataset and to explore potential patterns between menstrual cycle status and daily study behavior. Visual and descriptive analysis methods were used to examine variations in study duration across different conditions before applying formal statistical tests.

Study Duration Across Menstrual and Non-Menstrual Days

An initial comparison of daily study hours between menstrual period days and non-menstrual days was performed using boxplots. The distributions of study duration across the two groups showed substantial overlap, with similar median values. This visual observation suggested that menstrual period days may not be associated with a substantial reduction in daily study duration.

Temporal Trends in Study Behavior

To examine study behavior over time, daily study hours were visualized using time-series plots. Menstrual period days were highlighted to assess whether consistent drops in study duration coincided with these periods. The time-series analysis did not reveal a clear or recurring downward trend during menstrual days, indicating that study behavior remained relatively stable over time.

To reduce daily variability and better observe underlying trends, a 3-day moving average was applied to the study hour time series. The smoothed trend further supported the observation that study duration did not significantly decline during menstrual periods.

Study Duration by Cycle Day

Daily study hours were also examined by menstrual cycle day to investigate whether early menstrual days—often associated with stronger physical symptoms—were linked to reduced study focus. The analysis did not reveal a systematic decrease in study duration during the initial days of the menstrual cycle, and study hours appeared relatively consistent across cycle days.

Influence of Behavioral Factors

Additional exploratory analyses examined the relationship between study duration and behavioral factors such as exam periods, coffee consumption, and painkiller usage. Visual comparisons indicated that exam periods were associated with noticeably higher study durations, suggesting that academic pressure had a stronger influence on study behavior than menstrual cycle status. In contrast, coffee consumption and painkiller usage did not exhibit clear or consistent patterns in relation to study duration.

Summary of EDA Findings

Overall, the exploratory analyses suggested that menstrual cycle status alone does not strongly influence daily study duration. Instead, external academic factors—particularly exam periods—appeared to play a more prominent role in shaping study behavior. These observations informed the formulation of the statistical hypothesis tests and guided the subsequent machine learning analysis.

Hypothesis Testing

Following the exploratory data analysis, formal statistical hypothesis testing was conducted to examine whether menstrual period days are associated with a significant change in daily study duration.

Hypothesis Formulation

Based on prior assumptions regarding hormonal fluctuations, physical discomfort, and fatigue commonly experienced during menstruation, the following hypotheses were formulated:

- **Null Hypothesis (H_0):**
  Menstrual period days are associated with a significant reduction in daily study duration.
- **Alternative Hypothesis (H_1):**
  Menstrual period days are not associated with a significant reduction in daily study duration.

Choice of Statistical Test

The outcome variable, daily study duration, is a continuous variable, while the predictor variable, menstrual period status, is binary. Normality tests (Shapiro–Wilk) indicated that study duration data did not follow a normal distribution. As a result, a non-parametric statistical test was required.

The Mann–Whitney U test was selected to compare daily study hours between menstrual and non-menstrual days, as it does not assume normality and is appropriate for comparing two independent groups.

Test Results

The Mann–Whitney U test revealed no statistically significant difference in daily study duration between menstrual period days and non-menstrual days ($p = 0.451$). The effect size was extremely small, indicating that menstrual period status has a negligible impact on study duration.

Interpretation of Results

The results of the hypothesis test do not support the assumption that menstrual period days significantly reduce daily study duration. Despite common expectations regarding decreased productivity during menstruation, the statistical analysis suggests that study behavior remains largely consistent across menstrual and non-menstrual days within this dataset.

These findings are consistent with the patterns observed during exploratory data analysis and provide a statistical foundation for further predictive modeling using machine learning techniques.

Machine Learning Analysis

To examine and strengthen the findings obtained from exploratory data analysis and hypothesis testing, supervised machine learning methods were applied to assess whether menstrual cycle status can predict daily study duration. The machine learning analysis was designed as a complementary approach, focusing on predictive patterns rather than statistical significance.

Machine Learning Task Definition

The problem was formulated as a supervised regression task, where the objective was to predict daily study duration (in hours) based on menstrual cycle status and behavioral variables.

- **Target Variable:**
  Daily study duration (study_hours_daily)
- **Feature Variables:**
  Menstrual period status (is_period), exam period status (exam_period), coffee consumption (coffee_consumption), and painkiller usage (painkiller_usage)

Given the temporal structure of the dataset, preserving the chronological order of observations was essential for realistic model evaluation.

Model Evaluation Strategy

To prevent data leakage and ensure a realistic prediction scenario, a time-aware train–test split was employed. The dataset was divided chronologically into 80% training data and 20% testing data, without shuffling. This approach simulates real-world conditions by training models on past observations and evaluating their performance on future data.

Models Applied

The following supervised learning models, aligned with the course content, were implemented and compared:

1. **Multiple Linear Regression**
   Used as an interpretable baseline model to examine the direction and magnitude of each feature's effect on study duration.
2. **Decision Tree Regressor**
   Applied to capture potential non-linear relationships between menstrual cycle variables and study behavior.
3. **Random Forest Regressor**
   Used as an ensemble model to improve prediction stability and to obtain reliable feature importance estimates.

Evaluation Metrics

Model performance was evaluated on the test set using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2 Score which are standard regression metrics. These metrics allowed for a consistent comparison of predictive performance across all models.

Machine Learning Results

Across all models, predictive performance was limited, which is expected given the behavioral and highly variable nature of daily study habits. Among the applied models, the Random Forest Regressor achieved the lowest prediction error. However, R^2 values remained negative across models, indicating that daily study duration is influenced by factors beyond the variables included in the dataset.

Feature importance analysis from tree-based models revealed consistent patterns. Exam period status emerged as the dominant predictor of daily study duration, accounting for the majority of the model's decision-making process. In contrast, menstrual period status exhibited minimal predictive importance, suggesting a negligible effect on study duration.

Interpretation and Consistency with Previous Findings

The machine learning results align closely with both exploratory data analysis and statistical hypothesis testing. While menstrual cycle phases were initially expected to influence study behavior, machine learning models consistently indicated that menstrual period status does not meaningfully predict daily study duration. Instead, external academic pressures—particularly exam periods—play a substantially more influential role.

These findings demonstrate consistency across multiple analytical approaches and reinforce the conclusion that menstrual cycle status alone is not a primary determinant of study focus within this dataset.

Findings and Discussion

This study investigated the relationship between menstrual cycle phases and daily study duration using exploratory data analysis, statistical hypothesis testing, and supervised machine learning methods. Across all analytical stages, the findings consistently indicated that menstrual period status does not have a significant or meaningful effect on daily study duration within the scope of this dataset.

Exploratory data analysis revealed substantial overlap in study duration distributions between menstrual and non-menstrual days, with no clear visual evidence of reduced study activity during menstrual periods. Time-series analyses and rolling average trends further supported the observation that study behavior remained relatively stable across menstrual cycle phases.

Statistical hypothesis testing using the Mann–Whitney U test confirmed these visual findings, showing no statistically significant difference in daily study duration between menstrual and non-menstrual days. The extremely small effect size suggested that any potential impact of menstrual cycle status on study behavior was negligible.

Machine learning models provided additional support for these conclusions. While overall predictive performance was limited due to the inherently variable nature of daily study habits, feature importance analyses consistently identified exam period status as the dominant predictor of study duration. In contrast, menstrual period status contributed minimally to prediction performance across all models.

Taken together, these findings suggest that external academic pressures, such as exam periods, play a far more influential role in shaping study behavior than physiological menstrual cycle phases.

Importantly, the consistency of results across multiple analytical methods strengthens the reliability of the study’s conclusions.

Limitations and Future Work

Despite the strengths of this study, several limitations should be acknowledged. First, the analysis is based on a single-subject, self-tracked dataset, which limits the generalizability of the findings. While the longitudinal nature of the data provides detailed insight into individual study behavior, results cannot be assumed to represent broader populations.

Second, daily study duration was self-reported, which may introduce reporting bias or measurement inaccuracies. Additionally, the dataset did not include other potentially relevant factors such as sleep quality, stress levels, mood, or physical activity, all of which may influence cognitive focus and productivity.

Future research could address these limitations by incorporating larger and more diverse participant samples, collecting data across multiple menstrual cycles, and including additional behavioral and physiological variables. Expanding the analysis to personalized or individualized predictive models may also provide deeper insights into how menstrual cycle phases interact with study behavior on a case-by-case basis.

Overall, this study serves as an exploratory, data-driven examination of a commonly assumed relationship and highlights the importance of empirical analysis when evaluating physiological and behavioral interactions.